

Free Build & AI-Tooling Guide

How to implement the blockchain-C2 IoT-botnet detection project at zero cost, using open-source tools, free datasets, and AI coding assistants

1. Is it really free? — yes, essentially 100%

Every component of this project is free for academic use. There is no required paid cloud service, no paid dataset, and no proprietary software. The only real costs are your time and, optionally, a faster internet connection for downloading datasets. A mid-range laptop is enough; a free Google Colab or Kaggle notebook covers any GPU need for the optional deep model. The sections below list exactly what to use and where the AI assistants help most.

2. Free tools (open-source stack)

Stage	Free tool	Cost
Testbed / VM emulation	VirtualBox or QEMU; GNS3; OpenWrt device images	Free / open-source
Traffic capture	tcpdump, Wireshark	Free / open-source
Flow feature extraction	Zeek, CICFlowMeter, Scapy	Free / open-source
Data handling	Python, pandas, NumPy	Free / open-source
Classical ML	scikit-learn (Random Forest), XGBoost	Free / open-source
Deep model (optional)	PyTorch or TensorFlow/Keras (LSTM/CNN)	Free / open-source
Imbalance handling	imbalanced-learn (SMOTE), class weights	Free / open-source
Explainability	SHAP	Free / open-source
Plots / figures	Matplotlib, Seaborn	Free / open-source
Writing / LaTeX	Overleaf (free tier), or local TeX	Free tier
Version control	Git + GitHub (free)	Free

3. Free datasets (academic use)

Use these for the benign class, for realistic IoT noise, and for known-botnet comparison; generate the blockchain-C2 and relay traces yourself in the testbed.

Dataset	What it gives you
IoT-23 (Stratosphere/CTU)	Labelled benign + malicious IoT captures, incl. Mirai-family
Bot-IoT (UNSW)	Large IoT botnet traffic dataset with attack labels
N-BaIoT	Benign vs. botnet (Mirai/BASHLITE) per-device features
CIC-IDS / CSE-CIC-IDS	General benign + attack flows for the negative class
MedBloT / IoT-Flock (optional)	Additional benign IoT device traffic for realism

Your testbed traces	The novel blockchain-C2 + victim-relay behaviours (you create)
---------------------	--

Always check each dataset's licence/terms and cite it. Most are free for research with registration.

4. Free compute

- Local laptop: enough for feature extraction and the Random Forest / XGBoost models.
- Google Colab (free) or Kaggle Notebooks (free): GPU access for the optional LSTM/CNN, with no setup cost.
- GitHub Student Developer Pack (free if you have a student email): extra credits and tools, though not required for this project.

5. Using Claude, Codex/Copilot, and Gemini

AI coding assistants are genuinely useful here and all have free tiers. Think of them as a fast pair-programmer for the scripting-heavy stages, not as a replacement for understanding your own project. Below is where each helps most.

Project stage	How the AI assistant helps	Do it yourself
Testbed setup	Explain VM/network config; draft UPnP and resolver stubs	Actually building & isolating the lab; safe sample handling
Feature extraction	Write Zeek/Scapy/CICFlowMeter parsing scripts fast	Deciding which indicators matter (your novelty)
Model training	Draft scikit-learn/XGBoost/PyTorch training + CV loops	Interpreting results; defending choices to your guide
Ablation / SHAP / plots	Generate the evaluation and plotting boilerplate	Reading the plots and writing the analysis
Paper writing	Improve phrasing, structure, LaTeX formatting	The actual claims, results, and honesty about limits

5.1 Practical tips

- Rotate tools if you hit free-tier limits: use one for code drafting, another for debugging, another for writing. This effectively extends your free usage.
- Give the assistant your feature list and dataset schema up front — it produces far better code when it knows the exact columns and labels.
- Always run and test AI-generated code on a small sample before trusting it; ask the assistant to add comments so you can explain each step in your viva/defence.
- Never let AI invent results or citations. Every number in the paper must come from a run you actually executed, and every reference must be one you verified.

6. Honest caveats

Free to build — yes. Effortless — no. Two parts resist automation and are where your real effort goes: (1) the isolated testbed and safe generation of blockchain-C2 / relay traces, and (2) understanding your own results well enough to defend them. These are also exactly where your novelty and your marks come from,

so the effort is well spent. AI assistants accelerate the surrounding code, but the scientific judgement — which indicators, which experiments, what the results mean — must be yours.

Free-tier limits on AI tools change over time and heavy use may be throttled; plan to spread coding work across days or across tools. None of this blocks the project — it just means pacing.